

The Revised Simplex Method

The way we've performing the Simplex Method so far is by writing a full dictionary at each step, but this is potentially wasteful: the matrix formulas for the dictionary tells us that knowing the basic variables is enough to reconstruct the whole dictionary and we don't even need *all* of the dictionary to figure out what pivot to perform, and thus to figure out what the next basis will be. For example, we never need to know the current value of the objective function. From the z -row we only need the coefficients of non-basic variables, to pick the entering variable, and then to pick the exiting variables we only need two columns of the dictionary: we need the values of the basic variables in the solution associated to the dictionary and we need the coefficients of the exiting variable in the formulas for the basic variables.

For example in the following dictionary we don't need any of the question marks to figure out that x_2 should enter and x_6 should exit:

$$\begin{aligned}x_3 &= 3 + ?x_1 - 4x_2 + ?x_5 \\x_4 &= 1 + ?x_1 + 2x_2 + ?x_5 \\x_6 &= 2 + ?x_1 - 3x_2 + ?x_5 \\z &= ? - 2x_1 + 4x_2 + 2x_5\end{aligned}$$

The idea of the Revised Simplex Method is to avoid having to compute the full dictionary after every pivot. Instead, we'll only keep track of some of the information, including the current basis, and use the matrix formulas to compute the portions of the dictionary we need.

We'll describe two versions of the Revised Simplex Method: one where we only keep track of the current basis variables, the current solution and compute everything else (to avoid computing A_B^{-1} , we'll solve systems of equations instead), and one where we additionally keep track of A_B^{-1} .

Revised Simplex Method, version 1

In this version, we keep track only of the basis $\mathbf{x}_B$ and the current solution¹, $\mathbf{x}_B^* = A_B^{-1}\mathbf{b}$. To deal with any occurrences of A_B^{-1} in the formulas, we'll solve systems of equations. When we say we "keep track" of some data, it means that when we start a pivoting step we will have that data available for the current basis, and also that we must compute that data for the next basis. Throughout we will need to refer to the matrix formulas for the dictionary:

$$\begin{aligned}\mathbf{x}_B &= A_B^{-1}\mathbf{b} - A_B^{-1}A_N\mathbf{x}_N \\z &= \mathbf{c}_B^\top A_B^{-1}\mathbf{b} + (\mathbf{c}_N^\top - \mathbf{c}_B^\top A_B^{-1}A_N)\mathbf{x}_N\end{aligned}$$

We're **given** $\mathbf{x}_B$ and the numbers $\mathbf{x}_B^*$. Now we must determine the entering variable. For that we need to calculate the coefficients of the non-basic variables in the z -row, i.e., $\mathbf{c}_N^\top - \mathbf{c}_B^\top A_B^{-1}A_N$. All of the pieces, $\mathbf{c}_B$, $\mathbf{c}_N$, A_B and A_N , can be found from the original LP (or more, precisely, from the augmented $\mathbf{c}^{\text{aug}}$ and A^{aug} you get after adding the slack variables). But we'd like to avoid calculating A_B^{-1} , so we'll get $\mathbf{y}^\top := \mathbf{c}_B^\top A_B^{-1}$ by solving the system of equations $A_B^\top \mathbf{y} = \mathbf{c}_B$.

Once we have $\mathbf{y}$ we can compute the vector of coefficients of $\mathbf{x}_N$ in the z -row as $\mathbf{c}_N^\top - \mathbf{y}^\top A_N$. We can then following the standard pivoting rules to choose the entering variable: we look for the largest positive coefficient, and in case of ties pick the one corresponding to the non-basic variable x_j with smallest j .

To choose the entering variable, we need $\mathbf{x}_B^*$ and the vector of coefficients of x_j in the top portion $\mathbf{c}_B$ (the $\mathbf{x}_B$ part). From the matrix formulation of the dictionary, this vector is $-A_B^{-1}A_j$ (recall that A_j is the column of A^{aug} corresponding to the variable x_j). Again, we avoid inverting A_B by solving a system of equations, namely, we get $\mathbf{d} := A_B^{-1}A_j$ by solving $A_B\mathbf{d} = A_j$.

Once we have $\mathbf{d}$, we can choose the exiting variable according to the standard pivoting rules: x_j starts at 0 and is allowed to increase as long as all the $\mathbf{x}_B$ stay ≥ 0 . So, the value of x_j in the solution associated to the *next* basis (in which x_j will be basic!), is given by $\max\{t \geq 0 : \mathbf{x}_B^* - t\mathbf{d} \geq 0\}$. Of course, there may be no maximum, that is, it might happen that $\mathbf{x}_B^* - t\mathbf{d} \geq 0$ for all $t \geq 0$; if so, that means the LP is unbounded and that $\mathbf{x}_N = 0$, $\mathbf{x}_B = \mathbf{x}_B^* - t\mathbf{d}$ is an unbounded family of feasible solutions.

If there is a maximum value of t , one or more of the entries of $\mathbf{x}_B^* - t\mathbf{d}$ becomes 0 (when t has the maximum value). Each entry of that vector corresponds to a basic variable; and we choose the exiting variable x_i to be the one whose entry becomes 0—or, if there are several, the one with smallest i .

It only remains to make sure we have the next-basis versions of all the data we gave ourselves. The new $\mathbf{x}_B$ is, of course, just obtained from the previous one by removing the exiting x_i and putting in the entering x_j . The new associated solution requires is also pretty easy: as we said above, the value x_j^* of x_j in the next dictionary is that maximum t , and the new values of $\mathbf{x}_B$ are given by $\mathbf{x}_B^* - t\mathbf{d}$ (again, for that maximum t). Note that the vector $\mathbf{x}_B^* - t\mathbf{d}$ includes a zero in the entry corresponding to the exiting variable x_i , all the other entries correspond to variables that are still basic in the next dictionary.

Recipe

Summarizing the above discussion, these are the steps you take given the list of basic variables $\mathbf{x}_B$ and their values in the associated solution, $\mathbf{x}_B^*$:

1. Find $\mathbf{y}$ by solving $A_B^T \mathbf{y} = \mathbf{c}_B$.
2. Compute $\mathbf{c}_N^T - \mathbf{y}^T A_N$, which is the vector of coefficients in the z -row. Use it to choose the entering variable x_j by looking for the largest positive entry. If all entries of the coefficient vector are non-positive, then stop: you have reached optimality!
3. Find $\mathbf{d}$ by solving $A_B \mathbf{d} = A_j$.
4. Find the maximum value of t such that $\mathbf{x}_B^* - t\mathbf{d} \geq 0$. From now on, let t be that maximum.
5. The exiting variable x_i will be the one (with smallest i , in case of ties) for which the corresponding entry of $\mathbf{x}_B^* - t\mathbf{d}$ is zero.
6. Find the values of the basic variables in the next dictionary:
 - For x_j the value will be that maximum t .
 - The other basic variables are the current ones, $\mathbf{x}_b$, but without x_i . Their values are $\mathbf{x}_B^* - t\mathbf{d}$ —this will include the value zero for x_i , which won't be basic anymore.

Revised Simplex Method, version 2

In this version, we additionally keep track of the matrix A_B^{-1} . If we have A_B^{-1} , the steps where we compute $\mathbf{y}$ and $\mathbf{d}$ can be done by simple matrix multiplication instead of solving a linear system. Of course, it's not all roses: we need to figure out how to compute the *next* A_B^{-1} , that is, the matrix $A_{B'}^{-1}$ where B' is the next basis.

To write down the A_B and $A_{B'}$ matrices you need to pick a specific order for the basic variables. Let's agree that the basis $\mathbf{x}_{B'}$ will have the same order as $\mathbf{x}_B$ with the exiting variable x_i replaced by the entering variable x_j . With that convention, if x_i occurs in position ℓ of $\mathbf{x}_B$, then $A_{B'}$ and A_B differ only in the ℓ -th column: A_B has A_i as ℓ -th column, while $A_{B'}$ has A_j .

This means that $A_{B'} = A_B E$ where E is the matrix whose ℓ -th column is $\mathbf{d} = A_B^{-1} A_j$ and otherwise the identity matrix. Why is that? Well, for any matrices M and N , if the columns of N are $\mathbf{v}_1, \dots, \mathbf{v}_m$, it follows that the columns of the product MN are the vectors $M\mathbf{v}_1, \dots, M\mathbf{v}_m$. In particular, if the columns of the identity matrix are $\mathbf{e}_1, \dots, \mathbf{e}_m$, then $M\mathbf{e}_i$ is simply the i -th column of the matrix M . For the matrix E , whose columns are $\mathbf{e}_1, \dots, \mathbf{e}_{\ell-1}, \mathbf{d}, \mathbf{e}_{\ell+1}, \dots, \mathbf{e}_m$ we see that the columns of $A_B E$ are the same as the columns of A_B except that the ℓ -th one is $A_B \mathbf{d} = A_j$; i.e., the columns of $A_B E$ are exactly those of $A_{B'}$!

From this we can easily state the *update rule* for A_B^{-1} , namely, $A_{B'}^{-1} = E^{-1} A_B^{-1}$. And this is extremely handy, because E^{-1} is very easy to calculate: namely, E^{-1} also agrees with the identity matrix except in the ℓ -th column, which is given by the following recipe: if $\mathbf{d}^\top = (d_1 \ d_2 \ \dots \ d_m)$, then the ℓ -th column of E^{-1} is given by

$$(-d_1/d_\ell \ -d_2/d_\ell \ \dots \ -d_{\ell-1}/d_\ell \ 1/d_\ell \ -d_{\ell+1}/d_\ell \ \dots \ -d_m/d_\ell)^\top.$$

Notice that the ℓ -th entry of that vector is the one on the diagonal of the matrix E^{-1} and does *not* follow the pattern of the others.

For example, if $E = \begin{pmatrix} 1 & p & 0 & 0 \\ 0 & q & 0 & 0 \\ 0 & r & 1 & 0 \\ 0 & s & 0 & 1 \end{pmatrix}$, then $E^{-1} = \begin{pmatrix} 1 & -p/q & 0 & 0 \\ 0 & 1/q & 0 & 0 \\ 0 & -r/q & 1 & 0 \\ 0 & -s/q & 0 & 1 \end{pmatrix}$.

This is easy enough to check by multiplying out. But for completeness's sake let's prove it:

Proposition. If E is a square matrix that only differs from the identity matrix in the ℓ -th column, where E has instead the column $(d_1 \ d_2 \ \dots \ d_m)^\top$, then E^{-1} also only differs from the identity matrix in the ℓ -th column which is $(-d_1/d_\ell \ -d_2/d_\ell \ \dots \ -d_{\ell-1}/d_\ell \ 1/d_\ell \ -d_{\ell+1}/d_\ell \ \dots \ -d_m/d_\ell)^\top$.

Proof. Let $\mathbf{e}_1, \dots, \mathbf{e}_m$ be the columns of the identity matrix. We know that the columns of E are given by $E\mathbf{e}_k = \mathbf{e}_k$ for $k \neq \ell$ and $E\mathbf{e}_\ell = \sum_{k=1}^m d_k \mathbf{e}_k$. Multiplying both sides of these equations by E^{-1} gives us, first, that $\mathbf{e}_k = E^{-1}\mathbf{e}_k$ for $k \neq \ell$, and second, that $\mathbf{e}_\ell = \sum_{k=1}^m d_k E^{-1}\mathbf{e}_k$. So we've already obtained that all columns of E^{-1} other than the ℓ -th are the same as those of the identity matrix. Plugging that into the last equation we get, $\mathbf{e}_\ell = \sum_{k \neq \ell} d_k \mathbf{e}_k + d_\ell E^{-1}\mathbf{e}_\ell$. Solving, we get that $E^{-1}\mathbf{e}_\ell = \sum_{k \neq \ell} (-d_k/d_\ell) \mathbf{e}_k + (1/d_\ell) \mathbf{e}_\ell$, as desired. ■

Recipe

To summarize, these are the steps you take given the list of basic variables $\mathbf{x}_B$, their values $\mathbf{x}_B^*$ in the associated solution, and the matrix A_B^{-1} :

1. Compute $\mathbf{c}_N^\top - \mathbf{c}^\top A_B^{-1} A_N$, which is the vector of coefficients in the z -row. Use it to choose the entering variable x_j by looking for the largest positive entry. If all entries of the coefficient vector are non-positive, then stop: you have reached optimality!
2. Compute $\mathbf{d} = A_B^{-1} A_j$.
3. Find the maximum value of t such that $\mathbf{x}_B^* - t\mathbf{d} \geq 0$. From now on, let t be that maximum.
4. The exiting variable x_i will be the one (with smallest i , in case of ties) for which the corresponding entry of $\mathbf{x}_B^* - t\mathbf{d}$ is zero.
5. Find the values of the basic variables in the next dictionary:
 - For x_j the value will be that maximum t .
 - The other basic variables are the current ones, $\mathbf{x}_b$, but without x_i . Their values are $\mathbf{x}_B^* - t\mathbf{d}$ —this will include the value zero for x_i , which won't be basic anymore.
6. Find the next A_B^{-1} : it is $E^{-1} A_B^{-1}$, where E is the matrix that differs from the identity matrix only in the column in the position of the entering/exiting variable, where E has the vector $\mathbf{d}$.

Invertibility of A_B

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The formula $A_{B'} = A_B E$ finally let's us repair a long standing omission: we haven't actually proved that A_B is invertible! To be clear, we're not saying that whenever you pick some columns of A^{aug} that form a square matrix that matrix is automatically invertible. What *is* true is that for any basis arising in the Simplex Method, the corresponding A_B will be invertible, as required for the matrix formulas for the dictionary to make sense!

At the very first step of the Simplex Method the basic variables are the slack variables and A_B is an identity matrix, which is, of course, invertible. Now, imagine that for some dictionary A_B is known to be invertible. Let's show the next matrix, namely $A_{B'} = A_B E$, is invertible too. It's enough to show E is invertible, since then $A_{B'}$ is a product of two invertible matrices and thus invertible itself.

So why *is* E invertible? Well, we found the inverse already! But for that inverse to make sense, we do need that $d_\ell \neq 0$ ². Alternatively, expanding by the ℓ -th row, we see that the determinant of E is exactly d_ℓ . At any rate, all we have left to show is that $d_\ell \neq 0$. This comes from the way pivots are chosen: d_ℓ is minus the coefficient of the exiting variable x_j in the formula for the entering variable x_i , so it has to be strictly positive (since the coefficient must be negative, otherwise x_i wouldn't be the exiting variable).

¹ The $\mathbf{x}_B^*$ are the values of the *basic* variables in the solution associated to the dictionary for the basis B . This determines the entire associated solution, because the non-basic variables have the value 0.

² A point which was glossed over in the proof of the proposition!

— [Omar Antolín Camarena](#)